



INNOMATICS[®]
RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

BIKE PRICE ANALYSIS: INSIGHTS FROM USED BIKE MARKET DATA

ABOUT ME

- I am Yallapragada Anusha.
- Passionate about Data Analysis, Python, and Data Visualization.
- Skilled in data collection, cleaning, visualization, and analysis.
- Eager to apply data-driven insights to solve real-world problems.
- Continuously improving skills in web scraping, exploratory data analysis, and predictive modeling.

PREFACE

- ❑ INTRODUCTION
- ❑ BUSINESS PROBLEM
- ❑ OBJECTIVES
- ❑ WEB SCRAPING
- ❑ DATASET OVERVIEW
- ❑ DATA QUALITY ASSESSMENT
- ❑ DATA CLEANING – DATA PREPROCESSING
- ❑ HANDLING MISSING VALUES
- ❑ DATA TRANSFORMATION & FINAL CLEANING
- ❑ NON VISUALIZATIONS
- ❑ DATA VISUALIZATIONS
- ❑ CONCLUSION
- ❑ EXPERIENCE AND CHALLENGES

INTRODUCTION

- The two-wheeler resale market is growing rapidly due to increasing demand for affordable personal transportation.
- Online platforms list thousands of used bikes with varying prices, conditions, and specifications.
- However, price variation depends on multiple factors such as model year, usage, engine capacity, ownership history, and location.
- This project performs **Exploratory Data Analysis (EDA)** on a used bike dataset to understand pricing patterns and key influencing factors.
- Real-world resale data often contains missing values, inconsistent formats, and noise, making preprocessing a critical step before analysis.
- By analyzing historical listings, the project aims to uncover market trends and provide data-backed insights into the used bike ecosystem.

BUSINESS PROBLEM

- The used two-wheeler resale market lacks standardized pricing, leading to large variations for similar bikes.
- Buyers often struggle to assess whether a listed bike price is fair, overpriced, or undervalued.
- Sellers rely on intuition rather than data-driven insights, which can result in incorrect pricing and delayed sales.
- Resale platforms handle large volumes of listings but lack analytical insights to highlight optimal price ranges.
- Inconsistent data formats, missing values, and unstructured information make analysis and decision-making challenging.
- Without proper analysis, stakeholders cannot clearly identify the key factors that influence resale value.
- This creates uncertainty and inefficiency in the used bike marketplace.

OBJECTIVES

- To clean and preprocess real-world used bike listing data.
- To analyze price distribution and ownership trends using univariate analysis.
- To study relationships between price and factors such as kilometers run, model year, and engine capacity.
- To generate actionable insights that support buyers, sellers, and business platforms in decision-making.

WEB SCRAPING

- An online used two-wheeler resale platform was selected as the data source.
- Browser **Developer Tools (Inspect Element)** were used to identify relevant HTML tags and page structure.
- **Python libraries such as Requests and BeautifulSoup** were used to send HTTP requests and extract listing data.
- Publicly available web pages were fetched and parsed to collect structured information.
- The extracted data was consolidated into a single dataset for further processing and analysis.
- **Exploratory Data Analysis (EDA)** was performed to identify pricing patterns, trends, and relationships among features.

DATASET

	Bike Name	Model Year	Price	Location	KMs Run	Color	Engine CC	No of Owners
0	Suzuki Avenis 125 Standard Edition	2022.0	Rs. 69,000	Mumbai, Maharashtra	9,500 kms	Grey	125 cc	Single Owner
1	Royal Enfield Classic 650	NaN	Rs. 4,05,000	Hyderabad, Telangana	1,000 kms	NaN	NaN	Single Owner
2	Royal Enfield Continental GT 650	2025.0	Rs. 4,20,000	Hadapsar, Pune, Maharashtra	1,000 kms	NaN	650 cc	Single Owner
3	Suzuki Gixxer SF Fi	2018.0	Rs. 65,000	Hyderabad, Telangana	16,500 kms	NaN	NaN	Two
4	Suzuki Gixxer 150	2018.0	Rs. 60,000	KR Puram, Bangalore, Karnataka	30,000 kms	NaN	NaN	Single Owner
...	...	...	...	...	...	...	...	...
2022	Royal Enfield Classic 350 2024	2024.0	Rs. 1,80,000	KR Puram, Bangalore, Karnataka	NaN	NaN	NaN	Single Owner
2023	Suzuki Access 125	2015.0	Rs. 28,000	Mangalore, Karnataka	53,000 kms	NaN	NaN	Single Owner
2024	Yamaha MT-15 Ver 2.0	2023.0	Rs. 1,50,000	Belgaum, Karnataka	9,414 kms	NaN	NaN	Single Owner
2025	Royal Enfield Himalayan	NaN	Rs. 1,30,000	Borivali, Mumbai, Maharashtra	7,900 kms	NaN	NaN	Single Owner
2026	Suzuki Gixxer 150	2021.0	Rs. 45,000	Faridabad, Haryana	50,000 kms	Grey	155 cc	Single Owner

2027 rows × 8 columns

DATASET OVERVIEW

- The initial dataset consists of 2027 used bike listings with 8 features.
- Key attributes include bike name, model year, price, location, kilometers run, engine capacity, color, and ownership details.
- The dataset contains a mix of numerical and categorical variables.
- Model years range from 1980 to 2026, indicating both old and recent bike listings.
- Several columns contained missing values and inconsistent formats, requiring preprocessing.

DATA QUALITY ASSESSMENT

- Price, kilometers run, and engine capacity were initially stored as text with units and symbols.
- Significant missing values were observed in Color and Engine CC columns.
- Minor missing values were found in Model Year, Price, KMs Run, and Ownership columns.
- Data types were inconsistent, limiting direct numerical analysis.

DATA CLEANING - PREPROCESSING

- Removed currency symbols, commas, and units from numerical columns.
- Converted Price, KMs Run into numeric data types.
- Invalid or non-numeric values were converted to missing values (NaN).
- Standardized ownership labels for consistency (e.g., Single → One Owner).

HANDLING MISSING VALUES

- Median imputation was used for Price, KMs Run, and Model Year.
- Median was chosen as it is robust to outliers in resale price data.
- Categorical missing values were handled using:
 - Mode imputation for ownership
 - “Not Mentioned” category for color
 - “Unknown” category for Engine CC
- This approach preserved dataset size and reduced bias.

DATA TRANSFORMATION & FINAL CLEANING

- Location column was split into City and State for regional analysis.
- State-wise analysis enabled geographical trend identification.
- Duplicate records were removed to avoid redundancy.
- Final cleaned dataset contains 1980 records with zero missing values.

	Bike Name	Model Year	Price	Location	KMs Run	Color	Engine CC	No of Owners	State
0	Suzuki Avenis 125 Standard Edition	2022	69000.0	Mumbai	9500.0	Grey	125	One Owner	Maharashtra
1	Royal Enfield Classic 650	2020	405000.0	Hyderabad	1000.0	Not Mentioned	Unknown	One Owner	Telangana
2	Royal Enfield Continental GT 650	2025	420000.0	Hadapsar, Pune	1000.0	Not Mentioned	650	One Owner	Maharashtra
3	Suzuki Gixxer SF Fi	2018	65000.0	Hyderabad	16500.0	Not Mentioned	Unknown	Two Owners	Telangana
4	Suzuki Gixxer 150	2018	60000.0	KR Puram, Bangalore	30000.0	Not Mentioned	Unknown	One Owner	Karnataka
...	...	...	...	...	...	...	...	...	...
2002	Royal Enfield Classic 350 2024	2024	180000.0	KR Puram, Bangalore	20500.0	Not Mentioned	Unknown	One Owner	Karnataka
2003	Suzuki Access 125	2015	28000.0	Mangalore	53000.0	Not Mentioned	Unknown	One Owner	Karnataka
2004	Yamaha MT-15 Ver 2.0	2023	150000.0	Belgaum	9414.0	Not Mentioned	Unknown	One Owner	Karnataka
2005	Royal Enfield Himalayan	2020	130000.0	Borivali, Mumbai	7900.0	Not Mentioned	Unknown	One Owner	Maharashtra
2006	Suzuki Gixxer 150	2021	45000.0	Faridabad	50000.0	Grey	155	One Owner	Haryana

1980 rows × 9 columns

NON-VISUALIZATION

UNIVARIATE ANALYSIS

```
df["No of Owners"].value_counts(normalize=True) * 100
```

```
No of Owners
One Owner      95.303030
Two Owners     3.989899
Three Owners   0.404040
Four Owners    0.303030
Name: proportion, dtype: float64
```

Majority of bikes are single-owner vehicles.

```
df["State"].value_counts().head(10)
```

```
State
Maharashtra    514
Karnataka      354
Tamil Nadu     225
Telangana      186
Uttar Pradesh  143
Delhi          85
Gujarat        79
West Bengal    73
Haryana        65
Kerala         41
Name: count, dtype: int64
```

Karnataka and Maharashtra have the highest number of listings.

BIVARIATE ANALYSIS

```
: df[["Price", "KMs Run", "Model Year"]].corr()
```

	Price	KMs Run	Model Year
Price	1.000000	-0.049999	0.372130
KMs Run	-0.049999	1.000000	-0.066261
Model Year	0.372130	-0.066261	1.000000

Price has a positive correlation with model year, indicating that newer and higher-capacity bikes tend to be priced higher. Price shows a weak negative correlation with kilometers run, suggesting that higher usage slightly reduces the price.

```
df.groupby("Model Year")["Price"].median()
```

Model Year	Price
1980	177500.0
1986	150000.0
1993	40000.0
1995	80000.0
1997	6000.0
1998	80000.0
1999	175000.0
2000	90000.0
2003	120000.0
2005	17500.0
2006	7000.0
2007	22500.0
2008	15000.0
2009	20000.0
2010	25000.0
2011	25000.0
2012	30000.0
2013	35000.0
2014	40000.0
2015	42000.0
2016	45000.0
2017	52400.0
2018	63000.0
2019	73500.0
2020	80000.0
2021	100000.0
2022	100000.0
2023	120000.0
2024	120000.0
2025	160000.0

Name: Price, dtype: float64

Median price analysis shows that older bikes have highly variable prices due to rarity and condition, while recent bikes (2016–2025) show a clear upward trend in median prices, reflecting the market value of modern models. Using median instead of mean reduces the impact of extreme outliers and provides a more realistic view of typical bike prices for each year.

```
df.groupby("State")["KMs Run"].agg(["count", "median"]).sort_values(by="median", ascending=False)
```

State	count	median
Goa	5	60000.0
Jammu Kashmir	2	31500.0
Orissa	27	31000.0
Bihar	13	30000.0
Gujarat	79	27600.0
Chhattisgarh	3	25000.0
Dadra Nagar Haveli	1	23400.0
Kerala	41	23242.0
Himachal Pradesh	3	23000.0
Tamil Nadu	225	23000.0
Delhi	85	22000.0
Andhra Pradesh	38	21750.0
Karnataka	354	21000.0

Telangana	186	20000.0
Maharashtra	514	20000.0
Uttar Pradesh	143	20000.0
Rajasthan	34	19100.0
Haryana	65	17500.0
West Bengal	73	17000.0
Madhya Pradesh	34	15700.0
Jharkhand	2	14600.0
Chandigarh	8	13700.0
Punjab	25	13000.0
Assam	8	12900.0
Arunachal Pradesh	1	9910.0
Pondicherry	3	9000.0
Utarakhand	8	5750.0

Median kilometers run varies by state, highlighting regional differences in bike usage, condition, and resale potential.

MULTIVARIATE ANALYSIS

```
df.groupby(["State", "No of Owners"])["Price"].median().unstack()
```

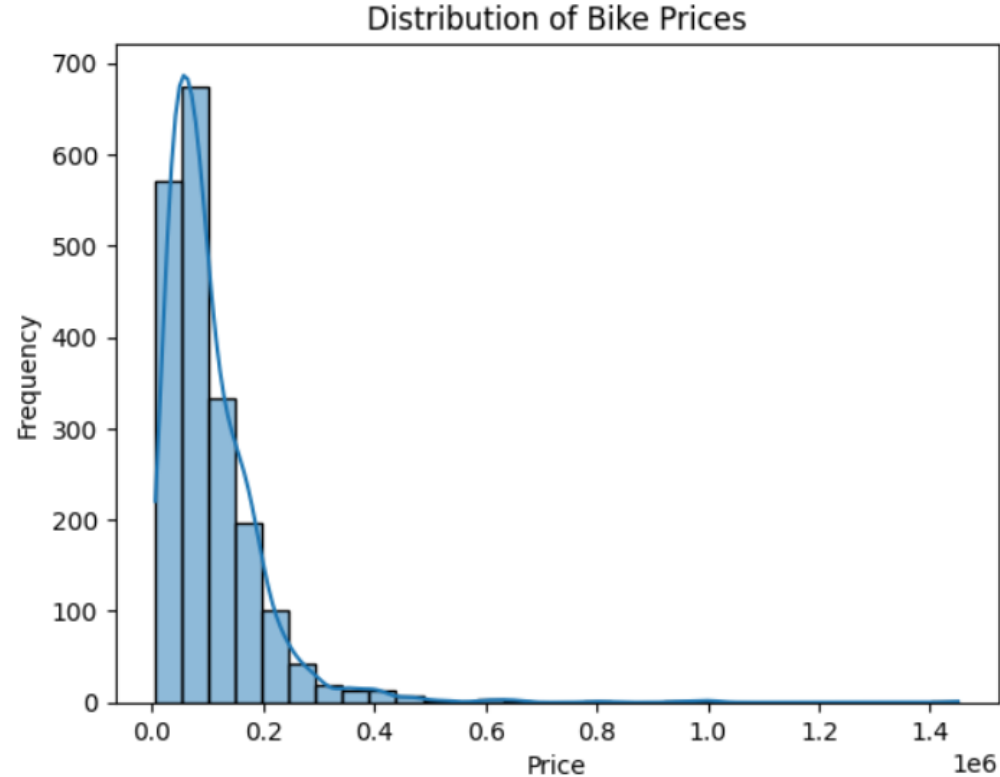
No of Owners	Four Owners	One Owner	Three Owners	Two Owners
State				
Andhra Pradesh	NaN	92500.0	NaN	NaN
Arunachal Pradesh	NaN	134000.0	NaN	NaN
Assam	NaN	167500.0	NaN	NaN
Bihar	NaN	52500.0	NaN	62000.0
Chandigarh	NaN	150000.0	NaN	NaN
Chhattisgarh	NaN	80000.0	NaN	NaN
Dadra Nagar Haveli	NaN	45000.0	NaN	NaN
Delhi	NaN	75000.0	25000.0	37000.0
Goa	NaN	49000.0	NaN	NaN
Gujarat	15000.0	67000.0	NaN	62500.0
Haryana	NaN	80000.0	NaN	72500.0
Himachal Pradesh	NaN	120000.0	NaN	NaN
Jammu Kashmir	NaN	130000.0	NaN	NaN
Jharkhand	NaN	130000.0	NaN	NaN

Karnataka	115000.0	90000.0	57500.0	75000.0
Kerala	20000.0	80000.0	NaN	NaN
Madhya Pradesh	NaN	100000.0	57000.0	88500.0
Maharashtra	600000.0	80000.0	59999.5	100000.0
Orissa	NaN	102500.0	NaN	65000.0
Pondicherry	NaN	100000.0	NaN	NaN
Punjab	NaN	120000.0	NaN	NaN
Rajasthan	NaN	80000.0	NaN	67000.0
Tamil Nadu	NaN	80000.0	NaN	67500.0
Telangana	NaN	89000.0	NaN	40000.0
Uttarakhand	NaN	179000.0	NaN	NaN
Uttar Pradesh	NaN	80000.0	45000.0	40000.0
West Bengal	NaN	70000.0	NaN	45000.0

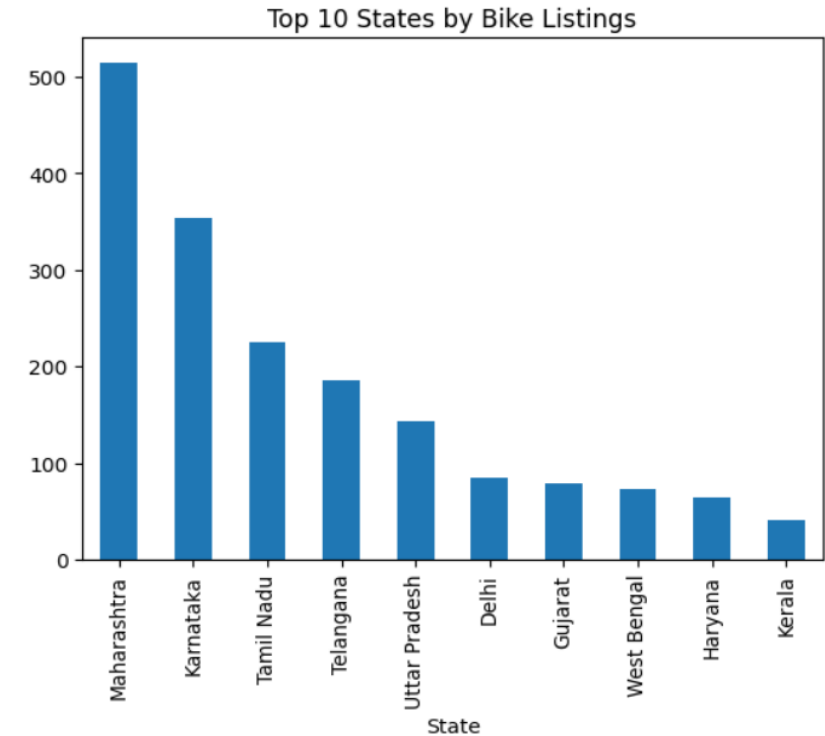
- Median bike prices vary across states and ownership categories, showing that single-owner bikes generally have higher prices while multi-owner bikes are priced lower.
- NaN values indicate the absence of listings for certain state–owner combinations rather than missing or incorrect data.

VISUALIZATION

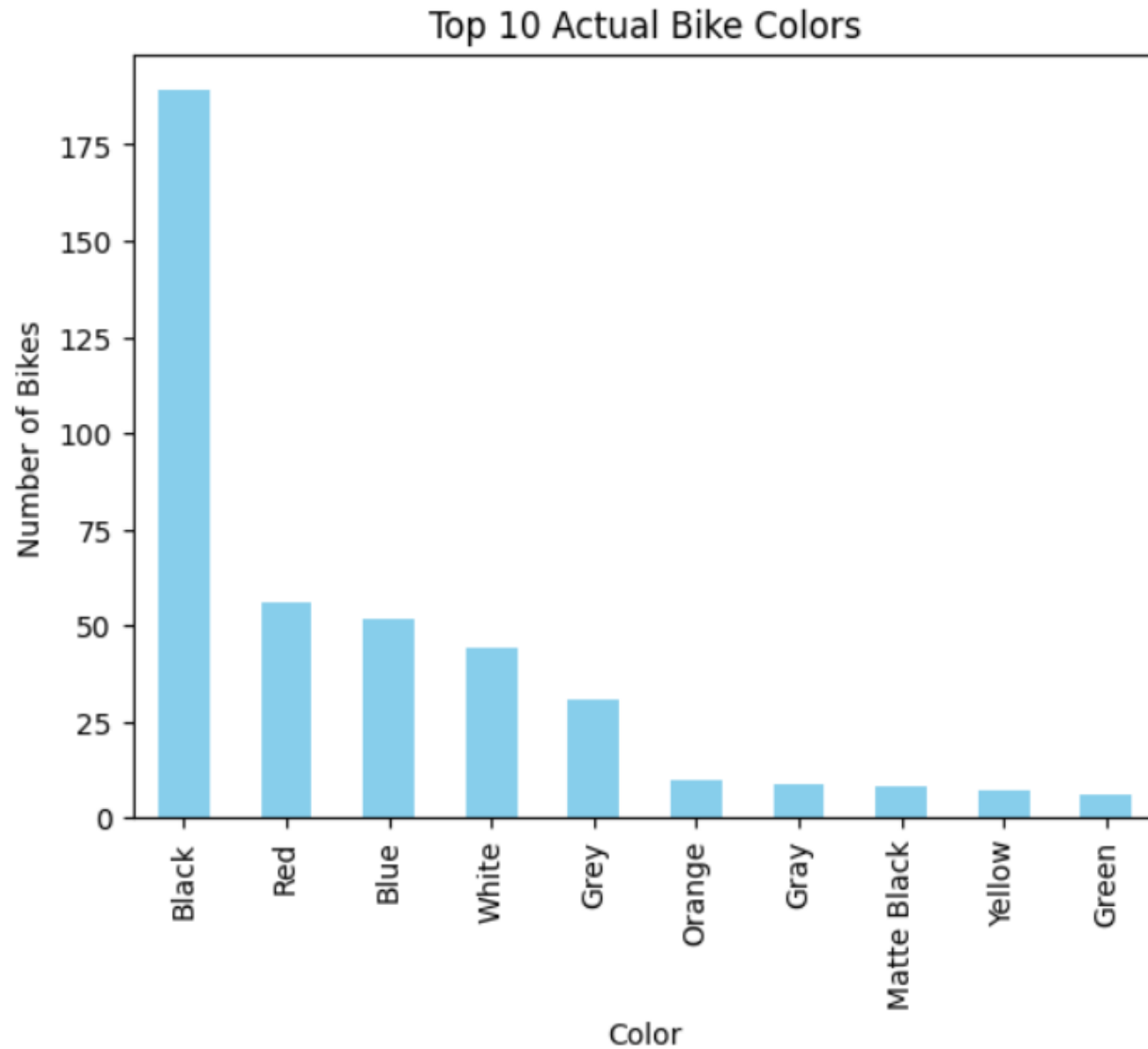
UNIVARIATE ANALYSIS



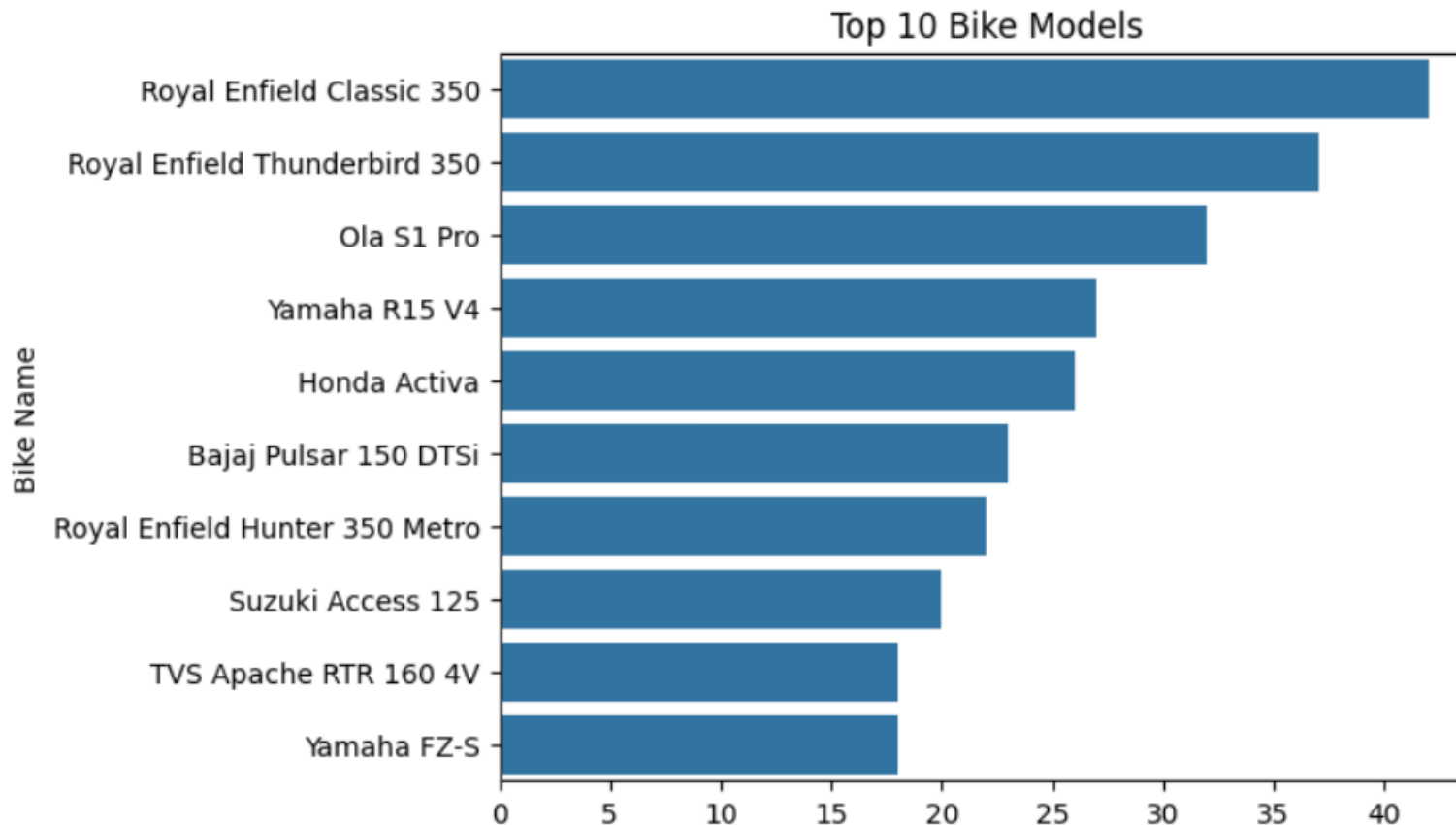
- Most used bikes are priced below ₹2 lakh, indicating a strong concentration in the affordable segment.
- A few high-priced bikes create a long right tail, showing that the price distribution is right-skewed due to premium models.



- The bar chart shows that Maharashtra has the highest number of bike listings, followed by Karnataka and Tamil Nadu. This indicates that bike listings are concentrated in a few major states, while other states have comparatively fewer listings.

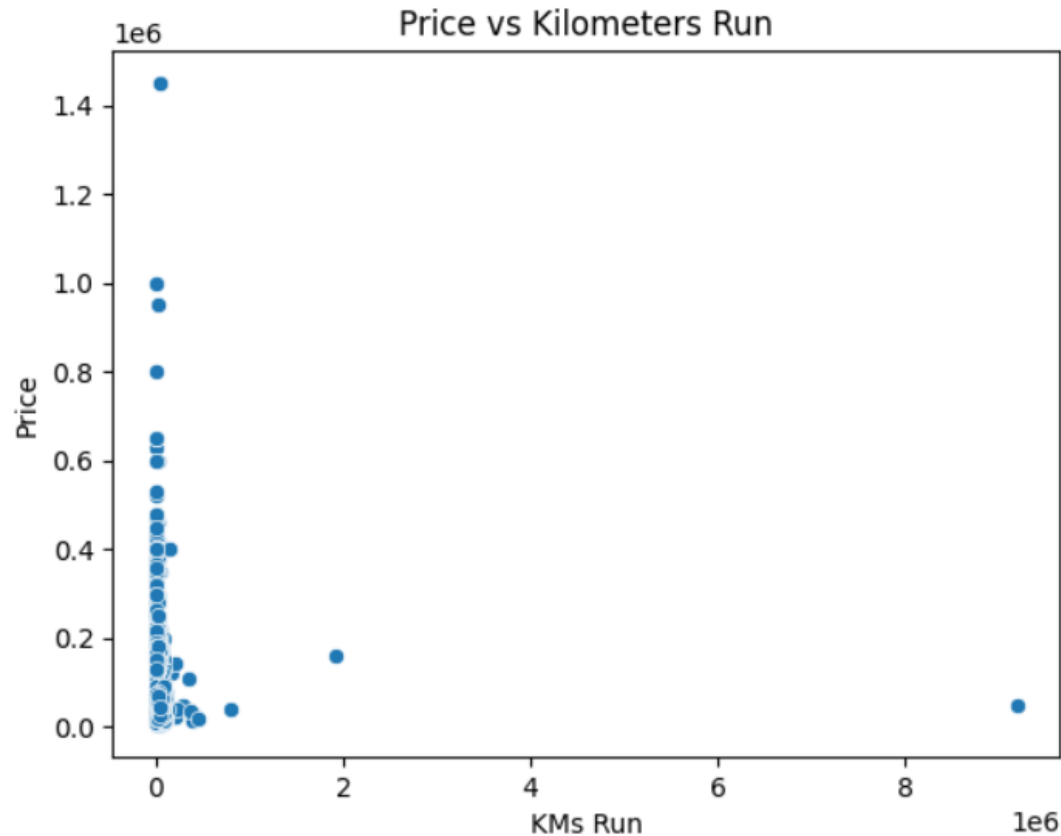


- Black is the most common color, dominating the market.
- Red, Blue, and White are moderately preferred.
- Less frequent colors include Grey, Orange, Green, and Yellow.
- Records with missing or unspecified color values were excluded.

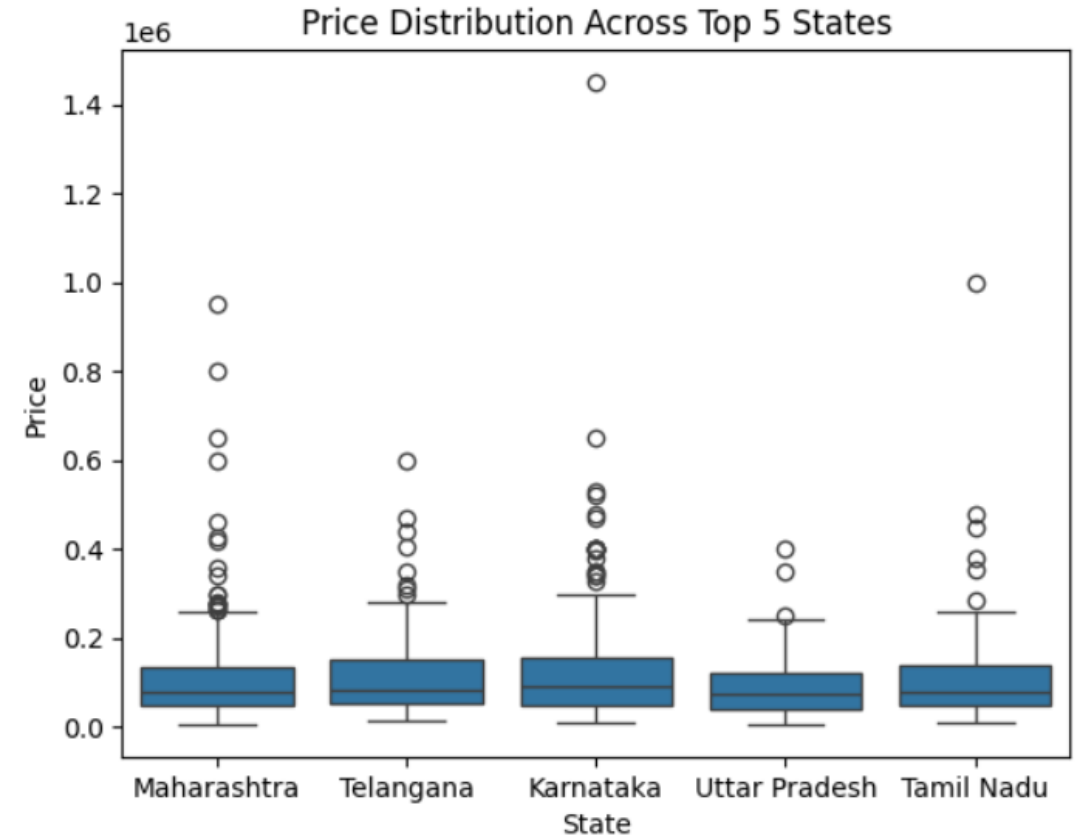


- Royal Enfield Classic 350 is the most frequently listed bike, followed by Royal Enfield Thunderbird 350.
- Popular commuter and affordable models like Ola S1 Pro, Yamaha R15, Honda Activa, and Bajaj Pulsar 150 are also among the top 10.
- This indicates strong resale demand for Royal Enfield bikes and high popularity of mid-range commuter bikes.

BIVARIATE ANALYSIS

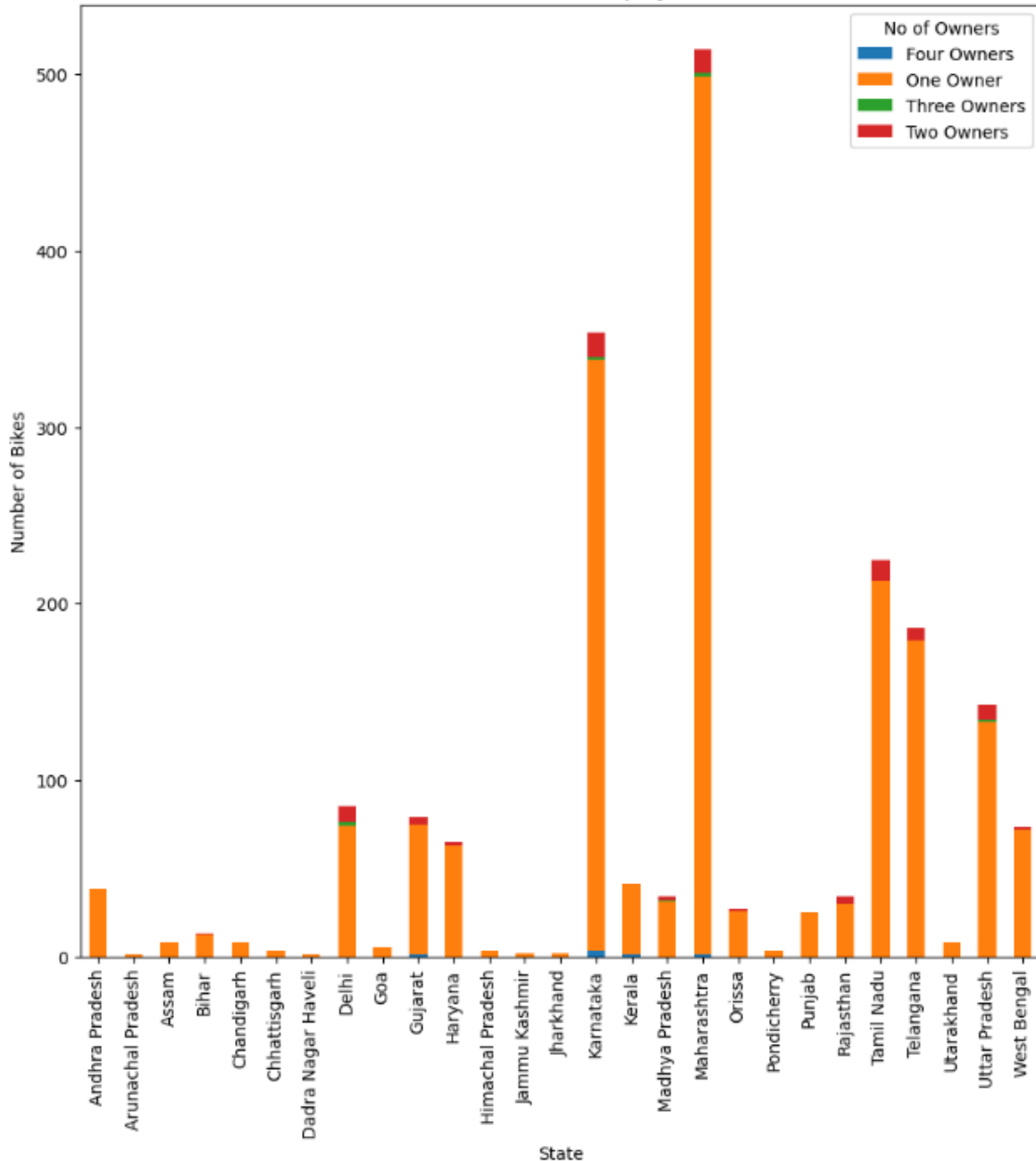


- There is a negative (inverse) relationship between kilometers run and bike price.
- Bikes with lower kilometers run tend to have higher prices, while prices decrease as kilometers increase.
- A few outliers exist with unusually high price or mileage.

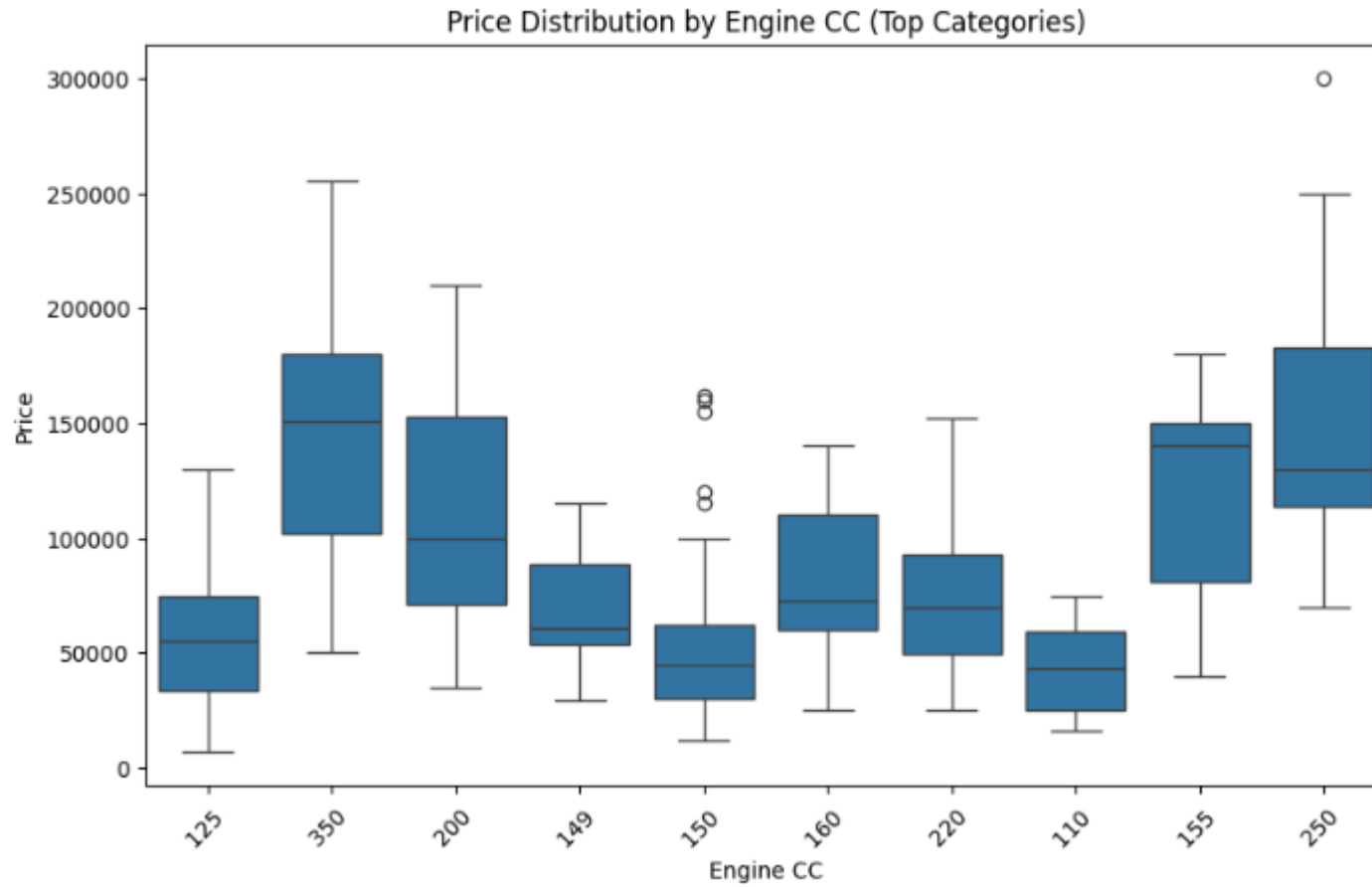


- Bike prices are unevenly distributed across states.
- States like Karnataka show wider price ranges and extreme outliers, indicating diverse market pricing.
- Other states have more compact distributions around the median, showing stable price trends.

Stacked Bar: Ownership by State

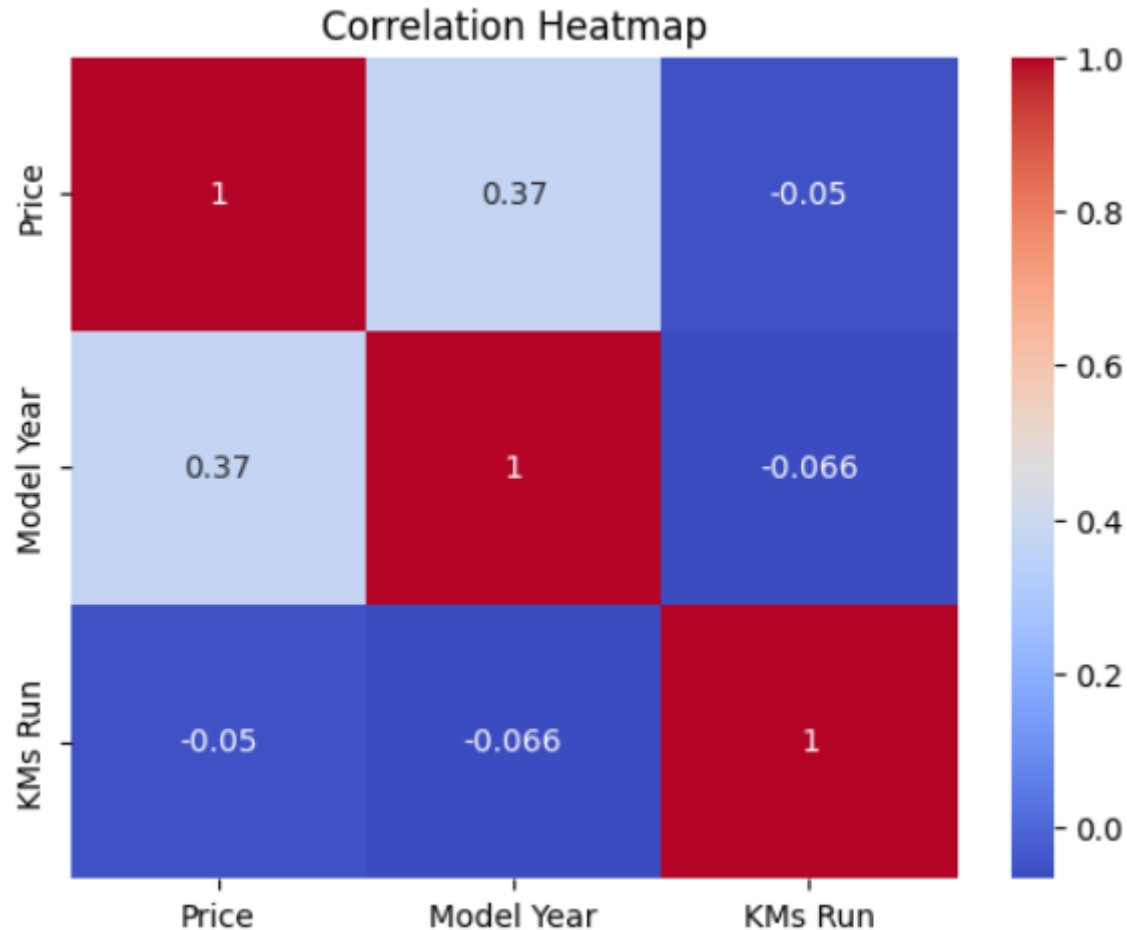


- Most bikes across all states are listed as “One Owner”, indicating high resale of relatively new bikes.
- Two, Three, and Four Owner bikes are comparatively rare in all states.
- States like Maharashtra, Karnataka, and Tamil Nadu have the highest total number of bikes, with the majority being one-owner listings.
- This suggests used bikes in popular states are mostly lightly used, making them attractive for resale.



- Higher engine CC bikes have higher median prices.
- Lower CC bikes are more affordable and price-stable.
- Price generally increases with engine capacity.

MULTIVARIATE ANALYSIS



- Price & Model Year: Moderate positive correlation (0.37), indicating that newer bikes generally have higher prices.
- Price & KMs Run: Very weak negative correlation, suggesting mileage has minimal impact on price.
- Model Year & KMs Run: Weak negative correlation, showing newer bikes tend to have slightly lower usage.
- Bike price is more influenced by model year than by kilometers run.

CONCLUSION

- Price trends: Most bikes are priced below ₹2 lakh; higher engine capacity and newer model year generally increase price.
- Bike preferences: Black is the most common color; Royal Enfield Classic 350 dominates the market.
- Ownership trends: Most bikes are sold by their first owner, with very few multi-owner bikes.
- Geographical trends: Prices vary across states, with Karnataka, Maharashtra, and Tamil Nadu showing wider price ranges.
- Influencing factors: Engine CC, model year, and brand impact price, but price is a multifactor decision.

EXPERIENCES AND CHALLENGES

Experiences:

- Extracted bike listings using Python (BeautifulSoup, Requests) efficiently.
- Gained hands-on experience in handling real-world messy datasets.
- Performed data cleaning, including removing units, handling missing values, and standardizing text.
- Learned to split, transform, and structure data for analysis (e.g., splitting Location into City & State).
- Explored trends in price, mileage, engine capacity, and ownership patterns.

Challenges:

- Dynamic web pages made scraping multiple listings challenging.
- Inconsistent data formats and missing values required careful cleaning.
- Numeric columns had special characters and units, needing preprocessing.
- Ensuring accuracy and consistency while handling duplicates and irregular entries.
- Managing large datasets and maintaining efficiency during scraping and cleaning.

THANK YOU

